Keyush Shah

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EDUCATION

University of Pennsylvania (School of Engineering and Applied Science) — **Philadelphia, PA**
*Master of Science in Engineering (**MSE**) in Data Science; **GPA 3.9/4.0*** — ***Aug 2023 – May 2025***
Coursework: *Machine Learning, Databases, Computer Systems, Deep Learning, Generative Modeling, NLP, Machine Perception*

EXPERIENCE

Software Engineer, Omnicom (NYC, New York) — ***Aug 2025 - Present***

- **Synthetic Audiences:**
  1. Architected a RAG-augmented product using **Next.js + FastAPI on Vercel/Railway** enabling pharma clients to simulate patient focus groups - interviewing AI-grounded personas & **reducing 2-4 weeks of research to minutes**.
  2. Researched and prototyped a statistical **Gaussian traversal** inference-time persona generator as a clustering-free alternative for data-scarce therapeutic domains – unlocking ad-hoc research via a **sub-second inference call**.
  3. Designed an HCP lens & ran a controlled A/B study -> **+10 pts on all 5 dimensions** vs patient lens (10x noise floor).
- **Brand Intelligence Compiler (Knowledge Base):**
  1. Engineered a Claude based multi-step agent that orchestrated **routing -> retrieval -> tool-use answering** over an internal pharma knowledge base that **eliminated fabricated claims** across 4 client projects.
  2. Designed an offline **eval harness** monitoring a north star GRR + diagnostic LLM judge flagging quality regressions.
- Led end-to-end migration of a multi-component ML Pipeline from AzureML to Databricks by re-architecting LLM Integration, data storage and maintaining orchestration in Spark Job DAGs unifying pipeline governance.
- Reduced CI/CD pipelines runtime by **12x** by migrating from Pipenv to **UV package manager**, configuring private package indexes and benchmarking Docker containers - packages installation **slashed to 12s across backend services**.

AI Researcher, *Penn Medicine* - *Computational Social Listening Lab* (PA, USA) — ***Dec 2024 – Dec 2025***

IH Risk Model ((Part Time Research) [CODE LINK]

- Reduced **model tuning time by 10x** by developing an **AutoML** pipeline using sklearn for **hernia** prediction **(recall: 0.9)**
- Deployed GPT-4 few-shot pipeline on 10k+ redacted surgical notes, **reducing 30% noise vs. BERT** for feature labeling.
- Built a fault tolerant multi-agent orchestration system using Claude Agent SDK featuring **LLM-as-judge** observability, automated retry logic and **FastAPI + WebSocket web interface** slashing pipeline runtime from **45 to 10 min.**
- Integrated vLLM & **LangChain** for scalable local LLM inference deployed on A100 GPU and **modularized the backend** (OpenAI, Mistral) using **parallelized batch inference** and **memory-aware chunking** across variable length notes.

Data Science Intern (Full-time), ***Universal Media*** (PA, USA) — ***May 2024 – Aug 2024***

- Led the development of **3+ data pipelines** using Azure Data Factory, facilitating seamless ingesting into Azure
- Drove product insights by building **Mixed Media time series models in Azure Synapse**, analyzing marketing channel impacts on media diversity metrics. Built **Power BI** dashboards to deliver actionable insights for optimizing client strategies.

Assistant Manager (Full-time), ***IIFL Finance Ltd*** — ***Apr 2022 – July 2023***

- **Digital Adoption:** Led a product initiative to identify digitally savvy customers **–** engineered features & built ADF pipelines to track campaign behavior. Trained an ML model in AzureML and exposed as a REST endpoint driving disbursal by ~50%.

SELECTED PROJECTS

- **Particle (2024):**
  - Built a context-aware chatbot for product and order support using FastAPI, LangChain, and Pinecone, implementing **RAG for product queries and tool-based logic with LLM fallback** for transactions.
  - Engineered multi-turn memory and entity tracking, **enabling >95% accurate follow-up handling**. Automated ingestion of synthetic data, & delivered **React + Vite frontend, improving support efficiency by 40%** and user satisfaction. [Link]
- **Ride Duration Prediction (2025):** Developed a production-ready ML pipeline to predict NYC taxi ride durations, using **Airflow** for orchestration and **MLflow** for experiment tracking. Achieved **~30% RMSE reduction** via automated hyperparameter tuning & designed modular, reproducible workflows to simulate real-world deployment as a web service via **Flask**. [Link]

TECHNICAL SKILLS

Programming Languages: Python, Go, C/C++, SQL, JavaScript, Typescript, R programming
ML Libraries/Frameworks: PyTorch, scikit-learn, XGBoost, LightGBM, HuggingFace, spaCy, OpenCV, MLflow, A/B Testing
Databases/Web Frameworks: MySQL, PostgreSQL, SSMS, MongoDB, Neo4j, React, NodeJS, Django, Flask, FastAPI
Cloud/Big Data Orchestration: AWS (S3, Glue), Azure (Data Factory, Synapse), GCP (BigQuery), Kafka, Airflow, DBT, PySpark
Tools/DevOps: DataBricks, Apache Spark, Docker, Jenkins, Git, Kubernetes, pytest, CI/CD